MIL-G-17586(NOrd)
2 July 1953

SUPERSEDING
NAVORD O.S. 1883

MILITARY SPECIFICATION
GAGE KIT, ELECTRICAL, FOR BORE AND RIFLING OF 20MM 9 GROOVE GUN BARRELS

1. SCOPE

1.1 This specification covers electric bore and rifling gage kits for ordnance use.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification and bidders and contractors should obtain the necessary copies:

SPECIFICATIONS

FEDERAL

NN-B-591 - Boxes, Wood-Cleated-Fiberboard.
NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 - Boxes, Fiber Corrugated (for Domestic shipment).

MILITARY

MIL-C-16232 - Coatings-Phosphate; oiled, slushed or waxed (For Ferrous Metal Surfaces) and Phosphate Treating Compounds.

NAVY DEPARTMENT

General Specifications for Inspection of Material and such parts of Appendix II as are applicable.

STANDARDS

MILITARY

MIL-STD-129 - Marking of Shipments.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following publications of the issue in effect on date of invitation for bids, forms a part of this specification:

NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw-Thread Standards for Federal Services

COMMERCE DEPARTMENT

CS8 - Gage Blanks

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Material.- The material shall be as specified herein.

3.2 The gage kit.- The gage kit shall consist of the following:

3.2.1 One (1) gage assembly for land diameter check.

3.2.2 One (1) gage assembly for groove diameter check.

3.2.3 One (1) gage assembly for groove depth check.

3.2.4 Three (3) extension rods.

3.2.5 One (1) set of eight (8) setting masters.

3.2.6 One (1) box for the gage assemblies, masters and extension rods.

3.2.7 One (1) portable instrument cabinet.

3.2.8 One (1) electric recorder with chain and reel.

3.2.9 One (1) constant voltage transformer.

3.3 Gaging depth.- Gages shall be capable of checking gun barrels to a depth of 72 inches from the muzzle end.

3.4 Range.- The gaging range shall be 0.030-inch.

3.5 Accuracy of gage readings.- Each gage assembly shall repeat readings to within 0.0002-inch between any 0.001-inch scale graduations and to within 0.0004-inch for full scale 0.030-inch deflection. The zero setting of the indicator shall not vary more than plus or minus 0.0005-inch for a plus or minus 20 degree

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Fahrenheit temperature variation from the original setting temperature condition.

The gaging points shall be so manufactured as to give accurate readings (within the above requirements) without the necessity of "rocking" the gage assembly in the gun barrel. This condition is required for all positions of the gage assembly in the barrel. The accuracy of the gage assembly shall not be affected by any position to which it may be rotated.

3.6 Gage assembly.- Each gage assembly shall consist of three parts: a gage head, an extension tube, and a handle.

3.6.1 Gage heads.-

3.6.1.1 Gage heads for land and groove diameters.- The gage head for land diameter and the gage head for groove diameter shall be a steel tube containing electrical coils, armature, movable carbide gaging point, fixed carbide gaging point and groove guide point. These points shall have spherical radii to avoid scoring the barrel and shall lie in a plane perpendicular to the axis of the gage head. This condition is stipulated to permit the use of these heads in gain twist gun barrels. The gage shall be capable of checking land diameters between 0.785-inch and 0.815-inch and groove diameters between 0.807-inch and 0.828-inch.

3.6.1.2 Gage head for groove depth.- The gage head for groove depth shall have four points. A movable indicating point shall contact the center of the bottom of the groove. Two stationary points shall contact the center of the lands adjacent to the groove being gaged. The fourth point to be diametrically opposite the indicating point, to be spring loaded and have a groove cut in its center portion so as to allow this point to span the land surface of the gun bore and rest on the bottom of the grooves adjacent to this land. The sides of this grooved point will bear on the sides of the gun grooves and serve to guide the gage head through the barrel. All points are to be in the same plane. The first three points mentioned should have sperical radii. The construction of this head, except for the points, shall be essentially the same as the head described in 3.6.1.1. The gage shall be capable of checking grooves with a depth of from 0.005-inch to 0.025-inch.

3.6.2 Extension tubes.- Each steel extension tube shall have an outside diameter of 5/8-inch maximum, shall have 1-inch and 1/4-inch engraved graduations and shall be graduated throughout its entire length. The 1-inch graduations shall circle the tube whereas 1/4-inch graduations shall be short lines. The graduation lines shall be 1/64-inch wide by 1/64-inch deep and shall be well filled with a hard paint, enamel, or lacquer of contrasting color. Each extension tube shall be phosphate coated in conformance with the applicable requirements of Specification MIL-C-16232. Each inch mark shall be engraved with its distance from the gaging point. There shall be two sets of lines and figures diametrically opposite each other to facilitate reading. The insulated wires joining the head and handle ends shall be enclosed in a plastic insulating tube to prevent deterioration of the wire insulation and consequent

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short circuits. The extension tube is to be secured to the handle in a semi-permanent joint that will only be unfastened for repair servicing. An adjustable depth stop that will permit the gage head to stop at the origin of the rifling shall be provided for each extension tube.

3.6.3 The handle.- The handle shall be equipped with a swivel device to permit the entire gage to follow the twist of the rifling without the necessity of the operator turning his gage supporting hand. The handle shall be knurled and provided with an adequate strain relief at the junction of the cable. Both the handle and the strain relief shall be cadmium or zinc coated.

3.6.3.1 Cable and attachment plug.- Each gage assembly shall be equipped with approximately 15-feet of flexible rubber-covered four-conductor cable, extending beyond the gage with a suitable four-prong attachment plug. The cable shall be equivalent to the type designated commercially as "S" or "SJ". An adequate length of this type of cable having a suitable attachment plug shall also be provided with the instrument cabinet, electric recorder, and constant voltage transformer.

3.7 Extension rod.- One steel extension rod, capable of screwing into the gage head up to 1-inch, shall be provided with each gage assembly. The extension rod shall have a swivel hook at one end capable of accommodating the chain that drives the electric recorder in 3.9. The extension rod shall be of such length that the swivel hook will extend 10-inches forward of the plane of the gaging points. An adjustable depth stop shall be provided with each extension rod. Each extension rod shall be phosphate coated in conformance with the applicable requirements of Specification MIL-C-16232.

3.8 Portable instrument cabinet.- The portable instrument cabinet shall contain the power unit, electrical magnification and zero adjusting rheostats, selector switch for groove, land and groove depth assemblies and the indicating meter. The meter shall be the light beam type having a linear scale graduated in thousandths. The scale shall read from minus 2 to 0 to plus 28 with even primary graduations numbered and odd secondary graduations not numbered. The zero setting for the land diameter shall be the 0.787-inch (minimum land) master ring. The zero setting for the groove diameter shall be at the 17th graduation on the scale. Both the land diameter, groove diameter and groove depth gage assemblies are operated alternately by the use of the selector switch. The instrument cabinet shall operate on 115 volt 60 cycle current. In operation the indicating meter is set with the master rings. Readings are given on the meter as the gage head progresses through the barrel. The meter shall be operated off the constant voltage transformer which shall provide an output voltage of 115 volts within plus or minus 1 percent accuracy. A selector switch to permit separate operation of the continuous recorder shall also be a part of the instrument cabinet. The instrument cabinet shall be equipped with a carrying handle.

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3.9 The electric recorder.- The electric recorder shall have an effective chart width of 11-inches. The full scale pen travel shall consume 4 1/2-seconds for 60-cycle operation. The calibrated accuracy shall be within plus or minus 1/4 of 1 per cent of scale span for 12 millivolt (mv.) input and within plus or minus 0.030 mv. for scale spans under 12 mv. input. The sensitivity shall be 1/4 of 1 percent of scale span for input in excess of 10 mv. and plus or minus 0.007 mv. for a scale span less than 10 mv. The record shall be made by an ink contrasting in color to the printed lines on the chart paper and at the same time both the ink and printed lines shall be suitable for blueprinting. The recorder shall operate from a 115 volt 60 cycle single phase alternating current power supply. The recorder shall be unaffected by vibration. It shall be driven by a flexible chain connected to the swivel hook of the gage assembly so that a direct ratio can be maintained between the length of barrel and the chart paper. The ratio of the length of barrel to the length of chart paper shall be 3 to 1. The chart paper calibration will be determined jointly by purchaser and supplier. (See 6.2). The flexible chain shall operate off a unit mounted directly on the continuous recorder. A mechanical linkage shall be provided to permit rewinding of the chain without regearing by hand. This same linkage shall also rewind the chart paper to its original starting position. An indexing device shall be provided to rapidly adjust the indicating pen to nine separate zero positions equally spaced on the chart paper. A window opening in the meter cabinet shall be provided to permit adding of pertinent information for each barrel inspected. The recorder shall be equipped with a carrying handle.

3.10 Master rings.

3.10.1 Master rings for land diameter check.- The three steel master rings for land diameter check shall be made in accordance with publication CS8 except for a clearance cut to clear the groove guide, and to class "X" tolerances. The rings shall be made and marked with the following gaging diameters 0.787-inch (for minimum land diameter), 0.789-inch (for maximum land diameter), and 0.815-inch (for full scale magnification check).

3.10.2 Master rings for groove diameter.- The three steel master rings for groove diameter check shall be made in accordance with publication CS8 and to class "X" tolerances. The rings shall be made and marked to the following gaging diameters 0.817-inch (for minimum groove diameter) 0.821-inch (for maximum groove diameter) and 0.807-inch for magnification check, the groove diameter gage shall be capable of checking diameters between 0.807-inch and 0.828-inch.

3.10.3 Master rings for groove depth.- The two steel master rings for groove depth check shall be made in accordance with publication CS8 except for the groove described below and to class "X" tolerances. The bottom of the groove shall be on the

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surface of a cylinder concentric with the inside diameter (I.D.) of the ring gage. One ring gage to have a 0.4105-inch groove radius and 0.787-inch (I.D.). The second ring gage to have a 0.4085-inch groove radius and 0.789-inch (I.D.). The first mentioned ring gage shall be marked 0.017-inch groove depth. The second ring gage to be marked 0.014-inch groove depth.

3.11 Gage kit box.- A suitable, well-constructed, wood or metal box of ample capacity to house the gage assemblies in 3.2.1, 3.2.2, and 3.2.3, the setting masters in 3.2.5, and the extension rods in 3.2.4, shall be provided with each gage kit. This box shall be equipped with adequate latches and handles suitable for convenient handling, shall be subdivided to insure safe storage for each item to be housed, and shall be permanently and legibly marked with the following information:

Gage, Electrical, For Bore and Rifling of 20MM 9 Groove Gun Barrels
Manufacturer's Model and Serial Number (if applicable)
Manufacturer's Name or Trade-Mark.

3.12 Voltage transformer. - A voltage transformer shall be provided with each gage kit to meet the requirements in 3.8.

3.13 Screw threads. - All threaded parts including screws, bolts, and nuts shall conform to the applicable requirements of National Bureau of Standards Handbook H-28, Screw-Thread Standards for Federal Services.

3.14 Spare Parts.- 60 Rolls of chart paper, 120 feet long and a replacement pen shall be provided with each electric recorder.

3.15 Marking.- Each component of the electrical bore and rifling gage kit shall be clearly stamped, etched or engraved with the manufacturer's name or with a trade-mark of such known character that the source of the manufacturer may be readily determined.

3.16 Workmanship.- Shall be first class in every respect and be in accordance with good commercial gage practice.

4. 4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection.- Unless otherwise specified in the contract or order, material purchased under this specification for the Bureau of Ordnance shall be inspected at the U. S. Naval Gun Factory, Washington 25, D. C. For naval purchases, the general inspection procedures shall be in conformance with General Specifications for Inspection of Material.

4.1.1 Visual and dimensional inspection.- All material shall be visually and dimensionally inspected to verify compliance with

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the requirements of this specification.

4.2 Service test.- Each component of the gage kit shall be subjected to tests under practical service conditions to determine the ease of operation, general serviceability, and suitability.

4.3 Test procedure.- Each gage kit shall be checked for calibrating accuracy against the master rings and for the requirements of 3.5 through 3.10.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation.- Unless otherwise specified in the contract or order, preservation shall be in accordance with the manufacturer's standard practice.

5.2 Packaging.- Unless otherwise specified in the contract or order, commercial packaging will be acceptable.

5.3 Packing.- Unless otherwise specified in the contract or order, the subject commodity shall be packed for domestic shipment in wood-cleated-fiberboard, wood-cleated-plywood, nailed wood, or wirebound wood boxes conforming to Federal Specifications NN-B-591, NN-B-601, NN-B-621 or NN-B-631, respectively.

5.4 Marking.- Unless otherwise specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Gage kits covered by this specification are to be used to measure the land (bore), groove diameters and groove depths of 20 mm (9 groove) gun barrels shown in Figure 1.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Place of inspection if different from 4.1
(c) Preservation, packaging and packing requirements if different from 5.1, 5.2 and 5.3 respectively
(d) Marking requirements if different from 5.4
(e) Chart paper calibrations (See 3.9)

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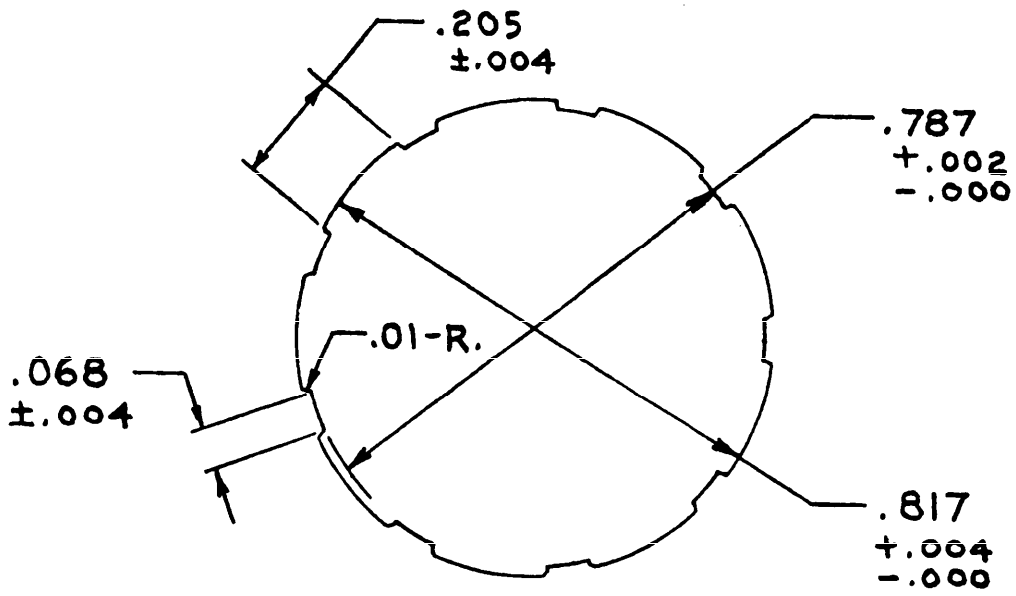


FIGURE 1. - 20 MM 9 GROOVE RIFLING

NAVY-OPPO PRNC WASH D.C

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